

N	TOPIC	PAGE	REMARKS
1.	Using the given data find the following data and make conclusion about the effect of outliers	1	} <u>4. interesting</u> 081-05-02
2.	Test which cricket team is more consistent using the given data	3	
3.	Compute central tendency, dispersion, skewness and kurtosis	4	} <u>4. interesting</u> 081-05-04
4.	Find the probabilities of asked problem based on given data	6	
5.	Find the probability that a student has a black coat from central region	7	} <u>4. interesting</u> 081-05-05
6.	Find the mean and standard deviation	8	
7.	What should be the price of fairy based on the probabilities	8	} <u>4. interesting</u> 081-05-06
8.	Find K , $P(0 < X < 0.5)$ and $E(X)$, from the given continuous RV	9	
9.	Test whether X & Y from joint distribution are independent or not	10	} <u>4. interesting</u> 081-05-07
10.	Find $V(X)$, $V(Y)$ and $COV(X, Y)$	11	
11.	Find the n to have probability to be 0.9	13	} <u>4. interesting</u> 081-05-08
12.	Find the probability of following problems	13	
13.	Find the probability of $(X \geq 3)$ and $(X = 5)$ using Poisson distribution	14	} <u>4. interesting</u> 081-05-09
14.	Find the probability if the lifetime of electronic component based on the condition.	14	
15.	Find the probability that a person have to wait at least 15 min to take a ride.	15	} <u>4. interesting</u> 081-05-11
16.	Find the probability that a printer can print more than 75 and less than 10 pages	15	
17. 18	Calculate the regression equation, find the asked values and interpret them	16, 18	

The following data represent the number of new computer account registered during 10 consecutive days.

- Compute mean, median, ~~mode~~, quartiles and sample st deviation
- Check for outliers using 1.5 IQR rule.
- Delete outliers and find data from (a), again
- Make a conclusion about effect of outliers.

X	43	37	50	52	58	105	52	45	45	10
---	----	----	----	----	----	-----	----	----	----	----

Actual formula used

$$\text{mean} = \text{SUM}(\text{datalist}) / \text{COUNT}(\text{datalist})$$

$$\text{Quartile}(n) = \text{QUARTILE}(\text{datalist}, n)$$

$$\text{Standard deviation} = \text{STDEV}(\text{datalist})$$

$$\text{IQR} = 1.5(Q_3 - Q_1)$$

$$\text{Upper limit} = Q_3 + \text{IQR}, \text{ Lower limit} = Q_1 - \text{IQR}$$

$$\text{Outlaw checker formula} = \text{IF}(\text{AND}(\text{Data} > \text{lower limit}, \text{Data} < \text{Upper limit}), "Not Outlaw", "Outlaw")$$

Formula with respect to sheet

With Outlaw one

$$\text{mean} = \text{SUM}(B2:B11) / \text{COUNT}(B2:B11), \text{ Median} = \text{QUARTILE}(B2:B11, 2)$$

$$Q_1 = \text{QUARTILE}(B2:B11, 1) \quad Q_3 = \text{QUARTILE}(B2:B11, 3)$$

$$\text{IQR} = 1.5 \times (D_4 - D_3)$$

$$\text{Upper limit} = (D_4 + D_5)$$

$$\text{Lower limit} = (D_3 - D_5)$$

$$\text{Standard deviation} = \text{STDEV}(B2:B11)$$

$$\text{Outlaw Condition} = \text{IF}(\text{AND}(B2 > D\$7, B2 < D\$6), "NOT", "OUTLAW")$$

AND SO, ON for rest.

Without Outlaw one.

$$\text{mean} = \text{SUM}(A15:A21) / \text{COUNT}(A14:A21), \text{ Median} = \text{QUARTILE}(A14:A21, 2)$$

$$Q_1 = \text{QUARTILE}(A14:A21, 1)$$

$$Q_3 = \text{QUARTILE}(A14:A21, 3)$$

$$\text{St. dev} = \text{STDEV}(A14:A21)$$

$$\text{Sorting formula} = \text{SORT}(A2:A11)$$

Excell sheet:

	A	B	C	D	E
1	X	sorted value			Outlaw finder
2	43	10	mean =	49.6	Outlaw
3	37	37	Q1 =	43.5	NOT
4	50	43	Q3 =	51.75	NOT
5	51	45	IQR =	17.375	NOT
6	58	45	Upper Limit =	64.125	NOT
7	105	50	Lower limit =	31.125	NOT
8	52	51	st. dev =	24.47	NOT
9	45	52	median =	47.5	NOT
10	45	58			NOT
11	10	105			Outlaw
12					
13	Writing X with Outlaws				
14	37	mean =	43		
15	43	median =	47.5		
16	45	Q1 =	44.5		
17	45	Q3 =	51.75		
18	50	st. dev =	6.46		
19	51				
20	52				
21	58				
22					
23	By removing those outliers we can see drastic change in				
24	mean and standard deviation while other data are				
25	relatively same.				

The number of run scored by Two cricket team are given in Test which are more consistent.

Group A	10	25	85	72	115	80	52	45	30	10
Group B	120	15	30	35	42	65	80	34	25	15

Formulas:

$$\text{Mean} = \frac{\sum x}{n}, \text{Stdev} = \text{STDEV}(\text{Data list})$$

$$= \text{SUM}(\text{data list}) / \text{COUNT}(\text{data})$$

$$\text{Coeff of variation} = (\sigma / \bar{x}) \times 100\%$$

Formula in reference to sheet

for A

$$\rightarrow \text{Mean} = \text{SUM}(A2:A11) / \text{COUNT}(A2:A11)$$

$$\rightarrow \text{Stdev} = \text{STDEV}(A2:A11)$$

$$\rightarrow \text{Coeff of variation} = (D4/D2) \times 100$$

Sheet.

for B

$$\text{Mean} = \text{SUM}(B2:B11) / \text{COUNT}(B2:B11)$$

$$\text{Stdev} = \text{STDEV}(B2:B11)$$

$$\text{Coeff of variation} = (D5/D3) \times 100$$

A	A	B	C	D
1	A	B		
2	10	120	Mean A =	52.4
3	25	15	Mean B =	46.7
4	85	30	St. dev of A =	35.01
5	72	35	St. dev of B =	33.16
6	115	42	Coeff of variation A =	66.83
7	80	65	Coeff of variation B =	71.9
8	52	80		
9	45	34		
10	30	25		
11	10	15		
12				
13	here coeff of variation of A < that of B, so, it mean A is more consistent.			

compute measure of central tendency, dispersion, skewness and kurtosis.

Wage	93-97	98-102	103-107	108-112	113-117	118-122	123-128	129-132
employees	2	5	12	17	14	6	3	1

Formula

Quartile class = $(N/4) \times n$, n = which class

Percentile class = $(N/100) \times n$, n = which one

Actual values = $L + \frac{H}{F} (\text{class no} - CF)$

Quartile deviation = $(Q_3 - Q_1)/2$

Bowleys skewness = $(Q_3 + Q_1 + 2MD)/(Q_3 - Q_1)$

kurtosis = $(Q_3 - Q_1)/(2 * (P_{90} - P_{10}))$

Formula on sheet

Median class = $E12/2$, Actual value = $C6 + ((D6 - C6)/E6) * (B14 - F5)$

And similarly for Q_1, Q_3, P_{10}, P_{90}

Quartile Deviation = $(F16 - F15)/2$

Bowleys skewness = $(F15 + F16 - 2 * F14)/(F16 - F15)$

Kurtosis = $(F16 - F15)/(2 * (F18 - F17))$

Sheet.

	A	B	C	D	E	F
	Wage	no.	making it	exclusive		
1					no. of employe	CF
2	lower	upper	lower	upper		
3	93	97	90.5	97.5	2	2
4	98	101	97.5	102.5	5	7
5	103	107	102.5	107.5	12	19
6	108	117	107.5	112.5	17	36
7	113	117	112.5	117.5	14	50
8	118	122	117.5	122.5	6	56
9	123	127	122.5	127.5	3	59
10	129	132	127.5	132.5	1	60
11				N =	60	
12	medial class =	30	107.5 to 112.5	class	Actual median =	110.74
13	Q1 class =	15	102.5 to 107.5	class	Actual =	105.83
14	Q3 class =	45	107.5 to 112.5	class	Actual =	115.71
15	P10 class =	6	97.5 to 102.5	class	Actual =	101.05
16	P90 class =	54	117.5 to 122.5	class	Actual =	120.83
17						
18	Quartile dev	4.82				
19	Bowley skewness	0.008	type skew			
20	kurtosis	0.256	leptokurtic			
21						
22						
23						

23. A company has made available a extensive health care club that may be used outside of work time. Records of last year indicate 250 employees, 110 used facility, 170 male were employed, 65 used

Find

- make a table
- What is probability that employee chosen at random is female and has used facility?
- Probability that employee chosen at random is a male.
- Probability that employee chosen at random is male or has not used the facility?

Sol'n

Probability formula = favourable event / Total event.

Excel sheet.

	A	B	C	D	E	F
1	Table					
2		Using facilities				
3		Yes	NO	P-Total		
4	male	65	105	170		
5	female	45	35	80		
6	P-Total	110	140	250		
7						
8	Probability the employee chosen at random is female and has used					
9	health facility =	0.18	formula = (B5/D6)			
10	probability that an employee chosen at random is					
11	a male =	0.68	Formula = D4/D6			
12	probability that employee chosen is a male or has not					
13	used health facility =	0.82	Formula = (B4/D6) + (C6/D6)			
14						
15						
16						

In a collage the geographical distribution of students. 50% from central, 30% from Eastern, 10% from Western, 8% from Mid-Western and 2% from Far-Western. The following proportion of student have black hair 80% of central, 70% of eastern, 60% western, 65% of midwestern and 75% of far Western.

find the probability that a student has a black hair come from central region.

Sol'n

formula $P(A|B) = P(A \cap B) / B$

Sheet:

	A	B	C	D	E	F
1	From central	Region (C) =	50%	0.5	$P(C)$	
2	From Eastern	Region (E) =	30%	0.3	$P(E)$	
3	From Western	Region (W) =	10%	0.1	$P(W)$	
4	From MidWestern	Region (M) =	8%	0.08	$P(M)$	
5	From Far Western	Region (F) =	2%	0.02	$P(F)$	
6						
7	Here B mean student with black hair.					
8	Black hair student in C=80%	$P(C B) =$	0.8			
9		$P(E B) =$	0.7			
10		$P(W B) =$	0.6			
11		$P(M B) =$	0.65			
12		$P(F B) =$	0.75			
13	Now					
14	$P(C \cap B) =$	0.4	Formula=	$D1 * D8$		
15	$P(E \cap B) =$	0.21	Formula=	$D2 * D9$		
16	$P(W \cap B) =$	0.06	Formula=	$D3 * D10$		
17	$P(M \cap B) =$	0.052	Formula=	$D4 * D11$		
18	$P(F \cap B) =$	0.015	Formula=	$D5 * D12$		
19	$P(B) =$	0.737	Formula=	$SUM(B14:B18)$		
20						
21	Now probability that student has a black hair is from central					
22	region	$P(B C) =$	0.543	$B14/B19$		
23						

5. Find the mean and standard deviation

$$\text{Formula } \bar{X} = \sum x \cdot P(x)$$

$$\sigma = \sqrt{\sum x^2 \cdot P(x) - (\sum x \cdot P(x))^2}$$

In Sheet.

	A	B	C	D	E	F
1	X	P(x)	X * P(x)	X ² * P(x)		
2	0	0.05	0	0		
3	1	0.35	0.35	0.35		
4	2	0.2	0.4	0.8		
5	3	0.25	0.75	2.25		
6	4	0.15	0.6	2.4		
7						
8	mean or Expected value		2.1	Formula =	SUM(C2:C6)	
9	Standard deviation =		1.179	Formula =	SQRT(SUM(D2:D6) - (C8^2))	

IF a man purchase a raffle ticket he can win 1st prize of 5000, 2nd of 3000, 3rd of 1000 with probabilities 0.001, 0.003 and 0.005 respectively. What should be fair price to pay for a ticket?
Sol'n

$$\text{Formula } \bar{X} = \sum x \cdot P(x)$$

Sheet.

	A	B	C	D	E
1	Prize list	Amount	Probability	$X \cdot P(X)$	
2	1st	5000	0.001	5	
3	2nd	3000	0.003	9	
4	3rd	1000	0.005	5	
5					
6	Expected value =	19	SUM(D2:D4)		
7	So, the prize of fair should be Rs. 19 for a ticket.				

12. A continuous RV x has following density function
 $f(x) = \begin{cases} kx^2 & \text{for } 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$

find (i) k . (ii) $P(0 < x < 0.5)$ (iii) $E(x)$

Soln, $f(x) = kx^2$ $[0, 1]$

a) For k , we know

$$\int_0^1 kx^2 dx = 1$$

$$\text{or } k \int_0^1 x^2 dx = 1$$

$$k \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$k \times \frac{1}{3} = 1$$

$$k = 3. \text{ So, } f(x) = 3x^2$$

$$\begin{aligned} \text{b) } P(0 < x < 0.5) &= \int_0^{0.5} 3x^2 dx \\ &= 3 \left[\frac{x^3}{3} \right]_0^{0.5} \\ &= 3 \left[\frac{0.5^3}{3} \right] \\ &= 0.125 \# \end{aligned}$$

c) $E(x)$ = mean, we know

$$E(x) = \int x \cdot f(x) dx$$

$$= \int_0^1 3x^3 dx$$

$$= 3 \left[\frac{x^4}{4} \right]_0^1$$

$$= 3 \times \frac{1}{4} = \frac{3}{4}$$

from given joint distribution. find

a) $P(X=1, Y \leq 1)$, b) $P(Y \leq 1)$ c) $P(X=2)$

$Y \backslash X$	-1	0	1
0	$1/15$	$2/15$	$1/15$
1	$3/15$	$2/15$	$1/15$
2	$2/15$	$1/15$	$2/15$

$$a) P(X=1, Y \leq 1) = P(X=1, Y=0) + P(X=1, Y=1) + P(X=1, Y=2) \\ = 1/15 + 1/15 = 2/15$$

$$b) P(Y \leq 1) = P(Y=0) + P(Y=1) \\ = (1/15 + 2/15 + 1/15) + (3/15 + 2/15 + 1/15) \\ = 2/3$$

$$c) P(X=2) = 1/15 + 3/15 + 2/15 = 6/15 = 2/5$$

11. Joint distribution of X and Y are given by $f(x, y)$. find
(i) conditional pdf X given Y (ii) test whether X & Y are independent
Given

$$f(x, y) = 4xye^{-(x^2+y^2)} \quad x \geq 0, y \geq 0$$

$$a) \text{ Marginal pdf of } Y = \int_0^{\infty} 4xye^{-(x^2+y^2)} dx$$

$$4ye^{y^2} \int_0^{\infty} x e^{-x^2} dx$$

$$\text{let } u = x^2, \quad du = 2x dx, \text{ so,}$$

$$4ye^{y^2} \int \frac{1}{2} e^{-u} du$$

$$\frac{4}{2} ye^{y^2} [-1 e^{-u}]_0^{\infty}$$

$$2ye^{y^2} [-1 e^{-x^2}]$$

$$-2ye^{y^2} \left[\frac{1}{e^0} + \frac{1}{e^{\infty}} \right]$$

$$= 2ye^{y^2} \cdot 1$$

$$= 2ye^{y^2}$$

Similarly for marginal pdf of $x = 2xe^{-x^2}$ #

conditional probability of y given x

$$f(y|x) = \frac{f(x,y)}{f(x)} = \frac{4xye^{-x^2-y^2}}{2xe^{-x^2}} = 2ye^{-y^2}$$

And

$$f(x|y) = 2xe^{-x^2}$$

To check if they are independent, $f(x,y)$ must be equal to $f(x) \cdot f(y)$

$$f(x,y) = f(x) \cdot f(y)$$

$$4xye^{-x^2-y^2} = 2xe^{-x^2} \cdot 2ye^{-y^2}$$

$$\text{or } 4xye^{-x^2-y^2} = 4xye^{-x^2-y^2}$$

so, they are independent.

Let (X,Y) be jointly distributed r.v. with pdf given $f(x,y)$
Find marginal pdf of X and Y. Find $V(X)$, $V(Y)$ and $\text{Cov}(XY)$

Given

$$f(x,y) = 2 \quad 0 < x < y < 1$$

here range is $(0 < x < y)$ and $(x < y < 1)$

a) Marginal pdf of X

$$f_X(x) = \int_x^1 f(x,y) dy$$

$$= \int_x^1 2 dy$$

$$= 2 \int_x^1 1 dy$$

$$= 2(y)_x^1$$

$$= 2(1-x)$$

Marginal pdf of Y

$$f(y) = \int_0^y f(x,y) dx$$

$$= 2 \int_0^y 1 dx$$

$$= 2(x)_0^y$$

$$= 2y$$

n) For $V(x) = E(x^2) - (E(x))^2$

So,

$$\begin{aligned} E(x) &= \int_0^1 x \cdot 2(1-x) dx \\ &= \int_0^1 2x - 2x^2 dx \\ &= \left[x^2 - \frac{2}{3}x^3 \right]_0^1 \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} E(x^2) &= \int_0^1 x^2 \cdot 2(1-x) dx \\ &= \int_0^1 2x^2 - 2x^3 dx \\ &= \left[\frac{2}{3}x^3 - \frac{2}{4}x^4 \right]_0^1 \\ &= \frac{1}{6} \end{aligned}$$

$$\begin{aligned} V(x) &= E(x^2) - (E(x))^2 \\ &= \frac{1}{6} - \left(\frac{1}{3}\right)^2 \\ &= \frac{1}{18} \end{aligned}$$

For $V(y) = E(y^2) - (E(y))^2$

So

$$\begin{aligned} E(y) &= \int_0^1 y \cdot 2y dy \\ &= \int_0^1 2y^2 dy \\ &= \left[\frac{2y^3}{3} \right]_0^1 \\ &= \frac{2}{3} \end{aligned}$$

$$\begin{aligned} E(y^2) &= \int_0^1 y^2 \cdot 2y dy \\ &= \int_0^1 2y^3 dy \\ &= \left[\frac{2y^4}{4} \right]_0^1 \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} V(y) &= E(y^2) - (E(y))^2 \\ &= \frac{1}{2} - \left(\frac{2}{3}\right)^2 \\ &= \frac{1}{18} \end{aligned}$$

c) For $\text{COV}(x, y) = E(xy) - E(x)E(y)$

$$\begin{aligned} E(xy) &= \int_0^1 \int_0^1 xy \cdot 2 dy dx \\ &= 2 \int_0^1 x dx \cdot \int_0^1 y dy \\ &= 2 \left[\frac{x^2}{2} \right]_0^1 \times \left[\frac{y^2}{2} \right]_0^1 \\ &= 2 \left[\frac{1}{2} \times \frac{1}{2} \right] = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{So, COV}(x, y) &= E(xy) - E(x) \cdot E(y) \\ &= \frac{1}{2} - \frac{1}{3} \times \frac{2}{3} \\ &= \frac{5}{18} \# \end{aligned}$$

The probability that a developer developing a software is $1/3$.
How many times must he try so that probability of developing is more than 90%.

Given
 $p = 1/3, q = 2/3, n = ?$

Since $X \sim B(n, p)$

$$P(X=x) = P(x) = {}^n C_x p^x q^{n-x} = {}^n C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{n-x}$$

$$x = 0, 1, 2, \dots, \infty$$

Here, the question

$P(x) = 0.9$, because, a person must have to develop either
So, $P(x \geq 1) = 0.9$ 1 or more to have that probability

$$\text{or } 0 - P(x=0) = 0.9$$

$$P(x=0) = 0.1$$

$${}^n C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{n-0} = 0.1$$

$$\Rightarrow \left(\frac{2}{3}\right)^n = 0.1, \text{ taking log}$$

$$n \log(2/3) = \log(0.1)$$

$$\Rightarrow n = 5.678 \sim 6, \text{ six developed software.}$$

77 The probability a student will graduate is 0.4. determine
probability out of 5 student (i) none (ii) atleast one (iii) all will

Given

$$p = 0.4, q = 0.6, n = 5$$

a) none will graduate, $x=0$.

$$P(x=0) = {}^5 C_0 (0.4)^0 (0.6)^{5-0} = 0.0776$$

b) Atleast one $P(x \geq 1)$

$$\begin{aligned} P(x \geq 1) &= 1 - P(x=0) = P(x=1) + P(x=2) + P(x=3) + P(x=4) + P(x=5) \\ &= 0.2592 + 0.3456 + 0.2304 + 0.0768 + 0.01024 \\ &= 0.9244 \end{aligned}$$

c) All will graduate $P(x=5)$

$$P(x=5) = {}^5 C_5 (0.4)^5 (0.6)^0 = 0.01024$$

d) Only one, $P(x=1)$

$$P(x=1) = {}^5 C_1 (0.4)^1 (0.6)^{5-1} = 0.2592$$

40. Messages arrive at an electronic message centre at random order, with average of 9 message per hour.

- Find the probability of receiving at least three message.
- Probability of receiving exactly five message

Formula:

Poisson distribution on Excel

= POISSON (data, lambda, bool)

bool for cumulative data.

In excel sheet.

	A	B	C	D
1	lambda =	9		
2	message at least than	3		
3	$P(X \geq 3) =$	0.979	Formula =	$1 - \text{POISSON}(B2, B1, \text{True})$
4	message is exactly	5		
5	$P(X = 5) =$	0.064	Formula	$\text{POISSON}(B4, B1, \text{False})$
6				

60. The lifetime of a certain electronic component with expectation of 5000 hrs and standard dev. 100. Compute, probabilities of

- life time is less than 5012
- Between 4000 to 6000 hrs
- more than 7000.

In excel sheet.

	A	B	C	D
1	Mean =	5000	St. dev =	100
2	less than 5012	X =	5012	
3	$P(X < 5012) =$	0.5478		
4	formula =	$\text{NORM.DIST}(C2, B1, D1, \text{True})$		
5	between 4000 + 6000	$X1, X2 =$	4000	6000
6	$P(4000 < X < 6000) =$	1		
7	formula =	$\text{NORM.DIST}(D4, B1, D1, \text{True}) - \text{NORM.DIST}(C4, B1, D1, \text{True})$		
8				
9	greater than 7000	X =	7000	
10	$P(X > 7000) =$	0		

74. Suppose that waiting time for a bus has a negative exponential distribution with parameter $\theta = 5 \text{ hr.}$. What is probability that a man has to wait at least 15 min? Also find expected time in

Formula in excel = EXPONDIST(x, θ , true or false)
In sheet.

	A	B	C	D	F
1	theta =	5	mean(1/theta) =	0.2	
2	Waiting time at least	15 min =	0.25 hr		
3	$P(x > 0.25) =$	0.287			
4	Formula =	1-EXPONDIST(0.25, 5, true)			
5					

76. A printer can print 15 pages per min. Use normal distribution as approx of Poisson distribution that printer can print
(i) more than 25 (ii) less than 10.

In Poisson distribution $\lambda = 1$
In sheet.

	A	B	C	D
1	mean =	St-dev =	15	
2	At least 25 page per minute, $x =$	25		
3	$P(x > 25) =$	0.742		
4	Formula =	1-NORM.DIST(25, 15, 15, true)		
5	less than 10 page per minute, $x =$	10		
6	$P(x > 10) =$	0.359		
7	Formula =	NORMDIST(10, 15, 15, true)		
8				
9				

A computer manager ~~instead~~ instead to know how efficient the new computer program which depend on size of incoming data. The data are given below for that purpose. Then find

- Interpret the regression coefficient with reference with problem.
- ~~Find~~ find coefficient of determination and interpret.
- Based on the model predict the efficiency of new computer for data size 12GB. Does it is possible to predict efficiency for data size of 30GB?

data size	6	7	7	8	10	10	15
Processor speed	40	55	50	41	17	20	16

Soln

Formula used

$$b_{yx} = \frac{n \sum xy - \sum x \cdot \sum y}{n \sum x^2 - (\sum x)^2}, \quad a = \frac{\sum y - b \sum x}{n}$$

coeff of determination = r^2

$$= \left(\frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \right)^2$$

Formula with respect to excel sheet.

a) $N = B10 = \text{COUNT}(B3:B9)$

(b) $X = A11 = \text{SUM}(A3:A9)$

c) $\sum x = B12 = \text{SUM}(B3:B9)$

(d) $\sum x^2 = D11 = \text{SUM}(C3:C9)$

e) $\sum y^2 = D12 = \text{SUM}(D3:D9)$

(f) $\sum xy = B13 = \text{SUM}(E3:E9)$

So

$$b_{yx} = ((B10 * B13 - B11 * B12) / (B10 * D11 - B11 * B11)) = B15$$

$$a = (B12 - B15 * B11) / B10 = B16$$

$$r = ((B10 * B13 - B11 * B12) / ((\text{SQRT}(B10 * D11 - B11 * B11)) * \text{SQRT}(B10 * D12 - B12 * B12)))) = B20$$

$$\text{coeff of determination} = B20 * B20 = B21$$

for Qn no. (C)

$$\text{When } x = 12, = B24, \quad Y = D24 = B16 + B15 * B24.$$

$$\text{When } x = 30 = B25, \quad Y = D25 = B16 + B15 * B25.$$

Excel sheet.

	A	B	C	D	E
1	Database (DB)	Process Request			
2	(X)	(Y)	X^2	Y^2	XY
3	6	40	36	1600	240
4	2	55	4	3025	110
5	2	50	4	2500	100
6	8	43	64	1849	344
7	10	17	100	289	170
8	10	26	100	676	260
9	15	36	225	1296	540
10	N=	7			
11	$\Sigma X =$	63	ΣX^2	673	
12	$\Sigma Y =$	205	ΣY^2	10079	
13	$\Sigma XY =$	1923			
14					
15	$b_{xy} = -4.143$	$b_{xy} = -4.143$ mean value of x change 7 unit			
16	$a = 22.286$	then y will change by 4.143 factor			
17	So, equation becomes as $Y = 22.286 - 4.143X$				
18					
19	Finding coefficient of determination $= r^2$				
20	$r =$	-0.87359			
21	coeff of det =	0.66			
22	So, this mean 66% on variation seen in y is caused by x, rest by other.				
23					
24	When x =	12	y =	22.577	
25	When x =	30	y =	-52	
26					
27	This equation can give data for x=12 but it can't be used in x=30				
28	because y is in Negative and Data request can't be negative and				
29	also x is outside of this model scope.				

A chemical company wishing to study the effect of extraction time on the efficiency of an extraction operation the data as.

- Fit a straight line to given line of find regression eqn and predict the extraction when time is 35 min
- Determine the coeff of determination and interpret.

Time	27	45	47	19	35	39	19
Efficiency	57	64	80	46	62	72	52

Sol'n

Formula used

$$b_{yx} = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2}, \quad a = \frac{\sum Y - b \sum X}{n}$$

$$\begin{aligned} \text{coeff of determination} &= (\text{correlation coeff})^2 \\ &= \left(\frac{n \sum XY - \sum X \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \sqrt{n \sum Y^2 - (\sum Y)^2}} \right)^2 \end{aligned}$$

Formula with respect to excel sheet.

- | | |
|------------------------------------|------------------------------------|
| (a) $\sum X^2 = A3:A9$ | (b) $\sum Y^2 = B3:B9$ |
| (c) $\sum X = A3$ | (d) $\sum Y = B3$ |
| (e) $\sum XY = A3:B3$ | (f) $n = \text{count}(B3:B9)$ |
| (g) $\sum X^2 = \text{SUM}(C3:C9)$ | (h) $\sum Y^2 = \text{SUM}(D3:D9)$ |
| (i) $\sum X = \text{SUM}(A3:A9)$ | (j) $\sum Y = \text{SUM}(B3:B9)$ |
| (k) $\sum XY = \text{SUM}(E3:E9)$ | |

AND here

$$B11 = \sum X, \quad B12 = \sum X^2, \quad D11 = \sum Y, \quad D12 = \sum Y^2$$

$$F11 = \sum XY$$

So,

$$b_{yx} = B13 = ((B10 * F11 - B11 * D11) / (B10 * B12 - B11 * B11))$$

$$a = B14 = (D11 - B13 * B11) / B10$$

For Qn(a), When $X = B18 = 35$, $Y = B14 + B13 * B18$

For Qn(b)

$$\text{Correlation coeff}(r) = C21 = ((B10 * F11 - B11 * D11)) / ((\text{SQRT}(B10 * B12 - B11 * B11) * \text{SQRT}(B10 * D12 - D11 * D11)))$$

$$\text{coeff of determination} = C22 = C21 * C21$$

Excell Table.

	A	B	C	D	E	F
1	Extraction(x)	Extraction(y)				
2	Time	Efficiency	X^2	Y^2	XY	
3	22	52	729	3249	1339	
4	45	64	2025	4096	2880	
5	45	80	1681	6400	3280	
6	19	46	361	2116	874	
7	35	62	1225	3844	2170	
8	35	72	1521	5184	2808	
9	19	52	361	2704	988	
10	$\Sigma X =$	275	$\Sigma X^2 =$	4333	$\Sigma XY =$	14539
11	$\Sigma X^2 =$	225	$\Sigma Y^2 =$	4333	ΣXY	14539
12	$\Sigma X^2 =$	7903	ΣY^2	27593		
13	$b_{yx} =$	0.9259				
14	$a =$	32.096				
15	So, this make the equation as $Y = 32.096 + 0.9259X$					
16	So this make the equation as $Y = 32.096 + 0.9259X$					
17	When $Y =$	35	$Y =$	64.503		
18	When $X =$	35	$Y =$	64.503		
19						
20	for coefficient of determination					
21	correlation coeff =	0.84				
22	coeff of determination =	0.71				
23	So this mean 71% of variation in Y is caused by					
24	variable X while rest caused by other variables.					
25						
26						

8. WAP TO HANDLE EXCEPTION FOR INVALID AGE INPUT

```
#include <iostream>
#include <stdexcept>
```

```
int GetAge() {
    int age;
    std::cout << "Enter age" << std::endl;
    std::cin >> age;
    if (age < 0 || age > 200) {
        throw std::out_of_range("Invalid age");
    }
    return age;
}
```

```
int main() {
    try {
        int age = GetAge();
        std::cout << "Your age is = " << age << std::endl;
    } catch (std::out_of_range e) {
        std::cout << "Error found = " << e.what() << std::endl;
    }
    return 0;
}
```

1) A researcher is studying the effectiveness of a new vaccine. They recorded 50 participants, with 4 virus each. The researcher recorded how many each participant remain unaffected. Fit it in Binomial Distribution.

Success	0	1	2	3	4
f	5	10	15	12	8

Formula used = $\text{BINOM.DIST}(x, n, p, \text{bool})$

	A	B	C	D	E
1	Success X	frequency f	fx		
2	0	5	0	Total trial =	200
3	1	10	10	Success trial =	108
4	2	15	30	Probability of P =	0.54
5	3	12	36	Probability of Q =	0.46
6	4	8	32	mean =	2.16
7	N =	50		n =	4
8	Formula used = $\text{BINOM.DIST}(x, n, p, \text{bool})$				
9	Binomial Probability	Binomial fit		Total formula =	50×4
10	0.04477	2.2387		Success =	Sum(fx)
11	0.21074	10.5122		P =	Success/Total
12	0.37071	18.5107			
13	0.28913	14.4866			
14	0.08503	4.2515			
15	Total =	50			
16					
17					

Fit the Poisson distribution on following dataset.

No of call	0	1	2	3	4	5	6
Frequency	2	3	5	4	3	2	1

Formula = POISSON(X, λ , bool)

	A	B	C	D	E
1	Num of call	Frequency	$f \cdot x$		
2	0	2	0	mean/lambda =	2.56
3	1	3	3	formula =	(9/189)
4	2	5	10		
5	3	4	12		
6	4	3	12		
7	5	2	10		
8	6	1	6		
9	Total =	20	53		
10	POISSON dist = POISSON(X, mean, bool)				
11	Poisson Prob	Poisson dist	Formula		
12	0.0707	1.47	$A12 * B\$9$		
13	0.1872	3.74	↓		
14	0.2483	4.96	↓		
15	0.2195	4.38	↓		
16	0.1452	2.90	↓		
17	0.0769	1.54	↓		
18	0.0340	0.68	↓		
19	Total =	19.62	↓		
20					

c) Fit normal dist in following dataset.

X	160-164	165-169	170-174	175-179	180-184
F	12 25	16 12	17 18	10 17	5 18

NORMAL (X, mean, std dev, bool)

	A	B	C	D	E	F
1		X				
2	lower	upper	M	f	Fm	1-Fm
3	160	164	162	5	810	131270
4	165	169	167	12	1004	334668
5	170	174	172	18	3096	537512
6	175	179	177	10	1770	313290
7	180	184	182	5	910	165620
8			Total =	50	8590	1427310
9	mean =	171.8	Formula =	E8/D8		
10	st. dev =	5.56	Formula =	sqrt((F8/D8) - B9*B9)		
11	Normal formula = (X, mean, std dev, bool)					

Normal 2

	lower	upper	Abs diff	Normal dist fit
15	0.012	0.080	0.0633	3.176
16	0.111	0.307	0.1966	9.879
17	0.373	0.654	0.2806	14.078
18	0.713	0.902	0.1848	9.239
19	0.930	0.986	0.0561	2.806
20			Total =	39.077